Structure of the heart

It is a muscular organ made up of smooth muscles (myocardium) which is enclosed in a sac known as the pericardium (outside layer). Shaped like an egg, is about the size of a person's clenched fist and weighs around 300 g in a man and 250 g in a woman.

The heart has four chambers, two atria (upper) and two ventricles (lower). Valves connect the upper and lower chambers and allows blood to flow only in one direction. The right and left sides of the heart are totally separated by a muscular wall and there is no communication between them.

The right side of the heart (Right atrium and Right ventricle) receives deoxygenated (impure) blood collected from different parts of the body through small and big veins (superior vena cava and inferior vena cava) which enters the lungs through the pulmonary artery.

In the lungs, blood is oxygenated and carbon dioxide and metabolic waste are removed. The left side of the heart (Left atrium and Left ventricle) gets oxygenated (pure) blood from the lungs through the pulmonary vein and supplies it to the entire body through the major blood vessel aorta and its innumerable branches (arteries and capillaries).

The left ventricle generates considerably greater pressure than the right ventricle, to enable the blood to be pumped throughout the body.

The heart muscles are supplied with oxygenated blood by two major coronary arteries and their branches which are distributed all over the surface of the heart.

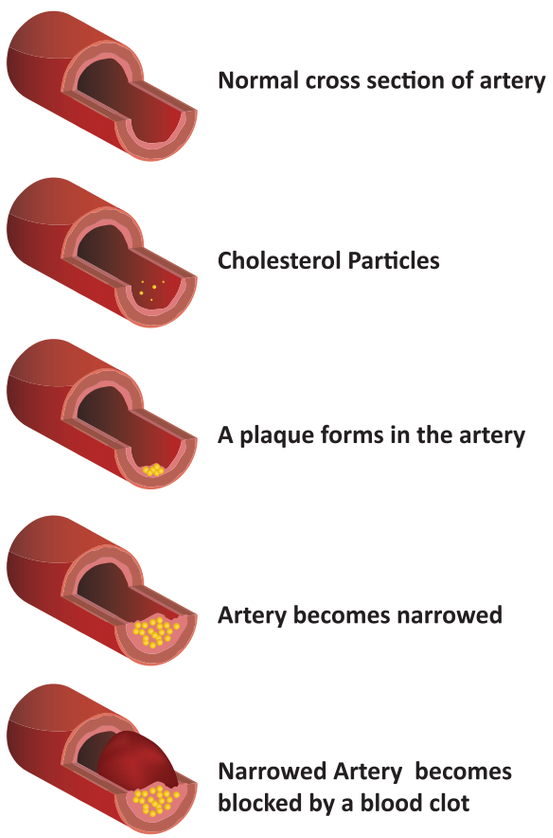
The coronary blood flow in a normal adult averages 200-250 ml/minutes. In a healthy adult at rest, the heart pumps approximately 5 litres of blood every minute. For efficient pumping, it is necessary for the heart to beat at a reasonable rate of 60-90 beats/minute which is achieved through controlled electrical impulses.

The heart and the blood vessels together constitute the cardiovascular system.

1.6.1 Cardiovascular diseases

The most common forms of cardiovascular disease are:

1. **Atherosclerosis** is a process by which arteries are narrowed due to the accumulation of cholesterol and other blood lipids along the walls obstructing the blood flow.



XII Std - Home Science EM_Unit 1.indd 20 20-04-2020 6.42.49 PM